

OSM / Opto-PING Rung 0N

K-Null Calibration Repair and Sensitivity-Specificity Balance v0.1

Status: PASS_NULL_CALIBRATION_REPAIRED

Executive summary

Rung 0N repairs the Rung 0M failure mode by making K significance conditional on a full reciprocal predictive win. With 60 positive-control trials and 350 null trials, positive full+K lock was 0.967, while null false full+K lock was 0.000. Raw null K significance was 0.011, but interpreted null K fell to 0.000 because K was not interpreted without full reciprocal prediction.

Locked specification

f11cc9662d5a666c10f29e11f70c88a04e377e6499e87ae0bb1f1632e884a6b8

Core metrics

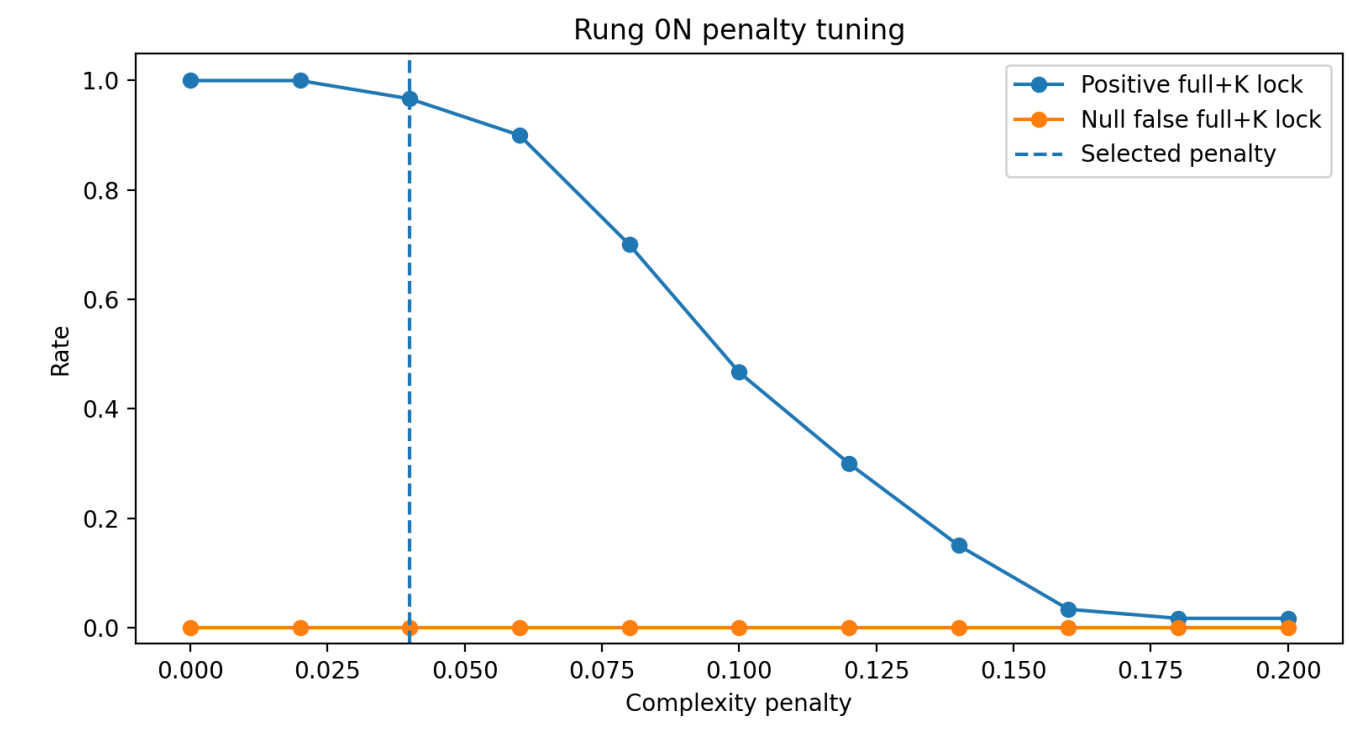
Metric	Result
Total trials	410.000
Positive trials	60.000
Null trials	350.000
Selected penalty	0.040
Positive full+K lock	0.967
Positive K significant	1.000
Positive K within 15%	1.000
Positive K within 10%	0.950
Null false full+K lock	0.000
Null raw K significant	0.011
Null interpreted K	0.000
Null non-full predictive win	1.000

Requested repairs implemented

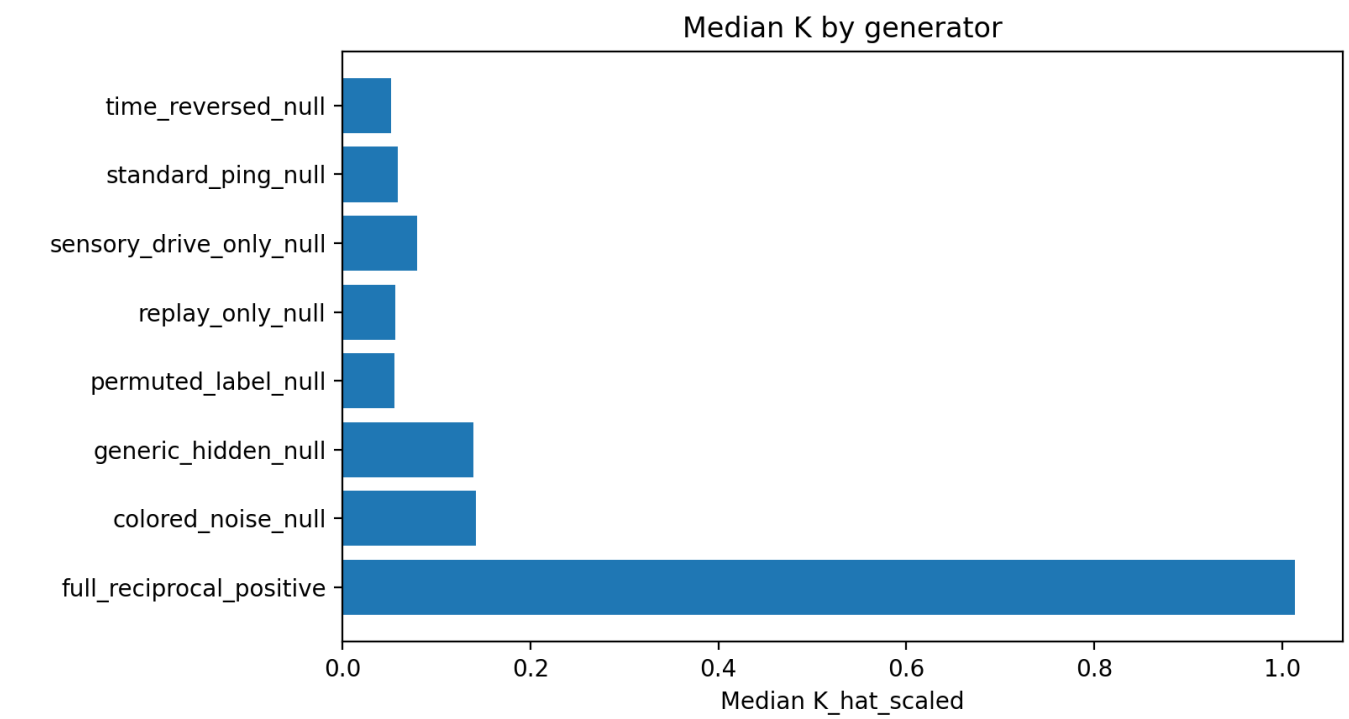
1. Higher-resolution null distributions: 5,000 calibration samples per null generator.
2. Generator-specific K thresholds: selected threshold = max(generator_q99, pooled_q975).
3. K is interpreted only when full reciprocal prediction also wins.
4. Penalty tuning restores positive sensitivity while retaining complexity penalty.
5. Strong colored-noise and generic-hidden alternatives remain in the model set.

Figures

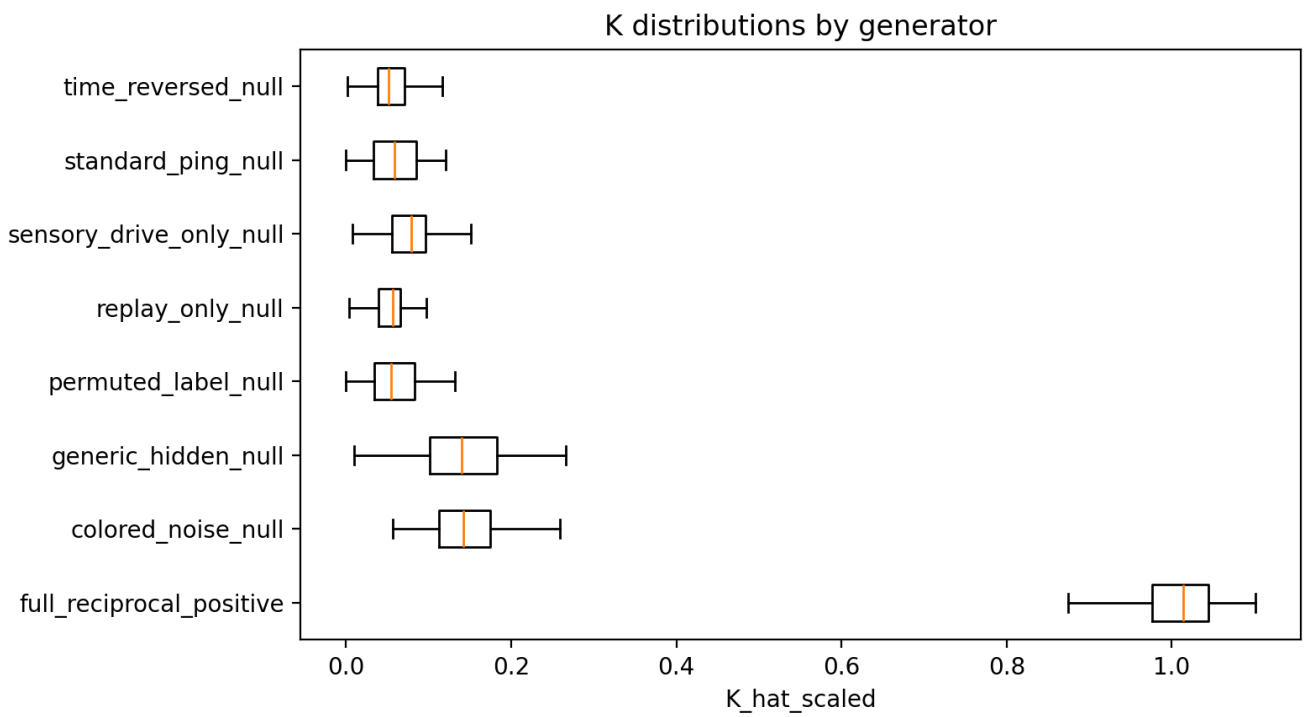
Penalty tuning: positive full+K lock versus null false lock.



Median K by generator.



K distributions by generator.



Generator summary

generator	generator_type	n_trials	median_K_hat_scaled	K_significant_raw_rate	full_predictive_win_rate	final_full_plus_K_lock_rate	dominant_model_mode
colored_noise_null	nan	50	0.142	0.020	0.000	0.000	sensory_drive_only
generic_hidden_null	nan	50	0.140	0.060	0.000	0.000	null_no_coupling
permuted_label_null	nan	50	0.055	0.000	0.000	0.000	sensory_drive_only
replay_only_null	nan	50	0.056	0.000	0.000	0.000	null_no_coupling
sensory_drive_only_null	nan	50	0.079	0.000	0.000	0.000	colored_noise_alt
standard_ping_null	nan	50	0.059	0.000	0.000	0.000	sensory_drive_only
time_reversed_null	nan	50	0.051	0.000	0.000	0.000	standard_ping
full_reciprocal_positive	positive	60	1.014	1.000	0.967	0.967	full_reciprocal

K-null calibration thresholds

generator	n_calibration	median_K_null	q99_K_null	selected_threshold
colored_noise_null	5000	0.134	0.342	0.342
generic_hidden_null	5000	0.123	0.274	0.274
permuted_label_null	5000	0.061	0.131	0.214
replay_only_null	5000	0.050	0.106	0.214
sensory_drive_only_null	5000	0.070	0.152	0.214
standard_ping_null	5000	0.055	0.124	0.214
time_reversed_null	5000	0.055	0.111	0.214

Penalty tuning

penalty	positive_full_plus_K_lock_rate	null_false_full_plus_K_lock_rate	positive_full_predictive_win_rate	null_full_predictive_win_rate	score
0.000	1.000	0.000	1.000	0.000	1.000
0.020	1.000	0.000	1.000	0.000	1.000
0.040	0.967	0.000	0.967	0.000	0.967
0.060	0.900	0.000	0.900	0.000	0.900
0.080	0.700	0.000	0.700	0.000	0.625
0.100	0.467	0.000	0.467	0.000	0.275
0.120	0.300	0.000	0.300	0.000	0.025
0.140	0.150	0.000	0.150	0.000	-0.200
0.160	0.033	0.000	0.033	0.000	-0.375
0.180	0.017	0.000	0.017	0.000	-0.400
0.200	0.017	0.000	0.017	0.000	-0.400

Lock decision table

criterion	value	target	pass
positive_full_plus_K_lock_rate >= 0.80	0.967	0.800	True
positive_K_significant_rate >= 0.90	1.000	0.900	True
positive_K_within_15_rate >= 0.90	1.000	0.900	True
null_false_full_plus_K_lock_rate <= 0.05	0.000	0.050	True
null_interpreted_K_rate <= 0.05	0.000	0.050	True
null_non_full_predictive_win_rate >= 0.85	1.000	0.850	True

Model-comparison fold winners

model	wins	total_folds	win_rate
sensory_drive_only	360	2460	0.146
zeta_only	327	2460	0.133
standard_ping	318	2460	0.129
full_reciprocal	312	2460	0.127
eta_only	311	2460	0.126
colored_noise_alt	264	2460	0.107
null_no_coupling	257	2460	0.104
generic_hidden_oscillator	172	2460	0.070
replay_only_basis	139	2460	0.057

Interpretive boundary

This remains synthetic sensitivity/specificity calibration only. It does not prove or disprove OSM, microtubular memory, biophotonic flickering, human EEG mechanism, or PR-AYC-G empirical results.

Next gate

Rung 0O should freeze the Rung 0N specification and run a fresh seed set without modifying thresholds, penalties, alternatives, or report template. After that, an independent verifier should run the unchanged package.